7.1.2. Binary Search Tree (BST) - Insert, Delete, and Search Operations

Code :

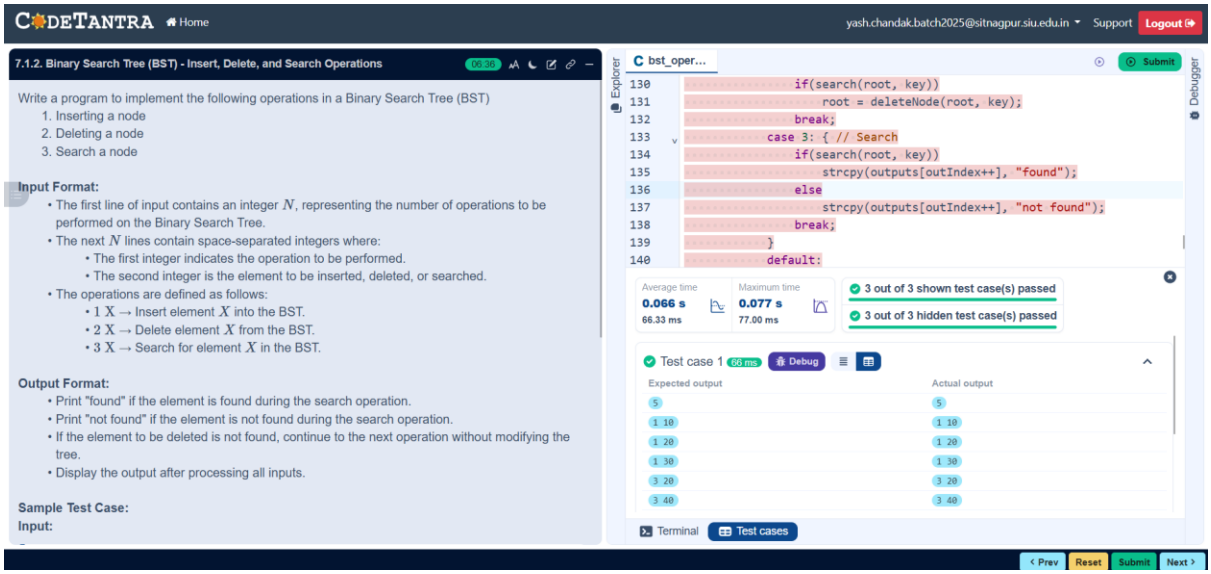
```
#include <stdio.h>

#include <stdlib.h>

#include <stdbool.h>

#include <string.h>



#define MAX_OPS 100



typedef struct Node {

 int data;

   struct Node* left;

   struct Node* right;



} Node;



Node* createNode(int data) {

           Node* newNode = (Node*)malloc(sizeof(Node));

   newNode->data = data;

   newNode->left = newNode->right = NULL;

   return newNode;



}
```

```
Node* insert(Node *root, int data) {
 if (root == NULL)
      return createNode(data);

   if (data < root->data)
      root->left = insert(root->left, data);
   else
      root->right = insert(root->right, data);

   return root;



}



Node* findMin(Node *root) {
  while (root && root->left != NULL)
      root = root->left;
   return root;









}
```

```
Node* deleteNode(Node *root, int key) {
 if (root == NULL)
      return NULL;

   if (key < root->data)
      root->left = deleteNode(root->left, key);
   else if (key > root->data)
      root->right = deleteNode(root->right, key);
   else {
      // node found

      // one or no child
      if (root->left == NULL) {
         Node* temp = root->right;
         free(root);
         return temp;
      }
      else if (root->right == NULL) {
         Node* temp = root->left;
         free(root);
         return temp;
      }

      // two children
      Node* temp = findMin(root->right);
      root->data = temp->data;
      root->right = deleteNode(root->right, temp->data);
```

```
    }
    return root;



}

// Search for a key in the BST.
bool search(Node *root, int key) {
 if (root == NULL)
        return false;

    if (key == root->data)
        return true;
    else if (key < root->data)
        return search(root->left, key);
    else
        return search(root->right, key);



}

int main() {
```

```
int opCount;
scanf("%d", &opCount);

Node *root = NULL;

char outputs[MAX_OPS][50];
int outIndex = 0;

int op, key;
for (int i = 0; i < opCount; i++) {
   scanf("%d", &op);
   if(op == 1 || op == 2 || op == 3)
      scanf("%d", &key);

   switch(op) {
      case 1: // Insert
         root = insert(root, key);
         break;
      case 2: // Delete
         if(search(root, key))
            root = deleteNode(root, key);
         break;
      case 3: { // Search
         if(search(root, key))
            strcpy(outputs[outIndex++], "found");
         else
            strcpy(outputs[outIndex++], "not found");
         break;
```

```
        }
        default:
            break;
    }
  }

  for ( int i = 0; i < outIndex; i++) {
printf("%s\n", outputs[i]);
        }
  return 0;
}
```

Programme :